SAIL TRIM TRAINING SIMULATOR

SailTrim Manual

How to use the simulator, the basics of trimming a mainsail and genoa, and what the simulator calculates behind the scenes.



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1 Quick start

SailTrim puts you on a sloop with a mainsail and a genoa. You steer and trim; the boat responds at once through its telltales, speed and heel. Nothing can break, so the fastest way to learn is to move one control and watch what changes.

Your first five minutes

1. **Look at the boat.** It starts on a beam reach with the wind from the side and both sheets half in. The **Trim** tile at the top shows how close you are to the best possible speed.
2. **Trim the genoa.** Drag the right-hand slider until both genoa telltales stream straight back. If the windward one lifts, sheet in. If the leeward one droops, ease.
3. **Trim the main.** Do the same with the left-hand slider and watch the two amber leech telltales.
4. **Read the hint.** The line above the rudder always names the single most useful thing to do next.
5. **Press Best trim.** Every sail control glides to the best setting for your present course and wind. Compare it with what you had.
6. **Change course and start again.** Head up towards the wind, then bear away, and re-trim each time. That is the whole game.

> **Tip.** Switch the wind to **manual** while you learn. A steady wind makes it far easier to see what each control does. Turn gusts and shifts back on once the basics feel natural.

What you are aiming for

- All telltales streaming, top and bottom, on both sails.
- Trim at 90 % or more (the number turns green).
- Moderate heel and only a few degrees of helm.

Where it runs

SailTrim is a single HTML file that runs in a browser on a tablet (touch, landscape) or a PC (keyboard and mouse). Settings are remembered by the browser on that device. On phones the screen is tight: some gauges and the on-screen wind panel are hidden, and you set the wind in Settings instead.

2 The screen



1. **Instruments.** Boat speed, target speed, trim %, wind and heel numbers. See section 4.
2. **Main sheet.** Up sheets in, down eases. The number is how far the sheet is pulled in.
3. **Genoa sheet.** Same, for the genoa.
4. **Telltales.** Genoa luff on the left, main leech on the right, each at the top and low down in the sail.
5. **Wind dial and heel.** Amber needle: apparent wind. Blue needle: true wind. The red and green arcs mark the close-hauled sector.
6. **Wind panel.** Switches between automatic and manual wind and, in manual, sets speed and direction.
7. **Traveller.** Moves the whole boom up to windward or down to leeward by up to 10°.
8. **Vang.** Holds the boom down so the main cannot twist open when the sheet is eased.
9. **Genoa car.** Forward closes the top of the genoa, aft opens it.
10. **Hint.** The most useful next action. Green dot: all good. Amber dot: something needs attention.
11. **Pause, Reset, Best trim.**
12. **Rudder.** Springs back to the centre when released. The small green tick shows the rudder needed to sail straight.
13. **Lock helm, view, settings.** Lock helm keeps the rudder where you leave it. The view button switches between heading-up and north-up.
14. **Boat and wind arrows.** Sails are drawn at their real angles; the dashed line is the upper part of each sail. A stalled sail turns pink. Blue arrow: true wind. Amber arrow: apparent wind.
15. **Help.** Opens this manual.

Dark patches moving across the water are gusts. Short streaks show the direction the true wind is blowing.

3 Controls and keys

Control	What it does	Range	Keys
Rudder	Steers. Returns to centre unless Lock helm is on.	30° each way	`A` `D` or `←` `→`
Main sheet	Sets how far the boom can swing out. Hard in, it also pulls the boom down and closes the leech.	boom 2° to 85°	`W` in, `S` ease
Traveller	Shifts the boom angle without changing twist.	10° up or down	`E` up, `Q` down
Vang	Limits main twist when the sheet is eased.	twist 24° down to 5°	`X` tighten, `Z` or `Y` ease
Genoa sheet	Sets the genoa angle. Easing also lets the top twist open.	11° to 60°	`↑` or `I` in, `↓` or `K` ease
Genoa car	Changes genoa twist. Shown as F5 (fully forward) to A5 (fully aft).	6° of twist each way	`U` forward, `O` aft

Button	What it does	Key
Best trim	Sets all five sail controls to the best trim for the present course and wind.	`T`
Wind: auto / manual	Switches between wind that varies by itself and wind you set.	`M`
Wind speed (manual)	One knot per step, 2 to 30 kn. Hold to repeat.	`−` `+`
Wind direction (manual)	Five degrees per step. Hold to repeat.	`,` `.`
Lock helm	Rudder stays where you leave it.	`L`
Heading up / North up	Changes how the view is oriented.	`V`
Pause	Freezes the boat. You can still move the controls.	`Space`
Reset	Back to the starting beam reach with default trim.	`R`
Help	Opens this manual in a new tab.	`H`

Steering notes

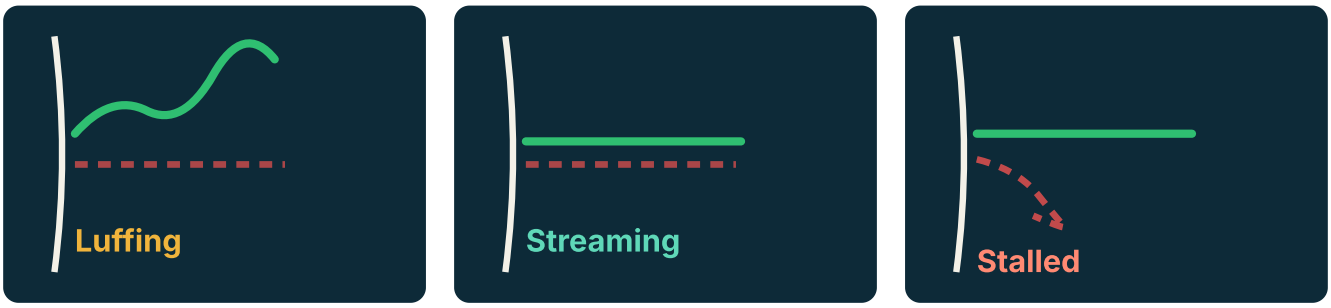
- **Port** and **Stbd** on the rudder slider are the direction the boat turns.
- The green tick on the rudder track marks the rudder angle that would hold a straight course. With the helm unlocked and released, the boat slowly turns the way its sails push it, usually up into the wind. Hold the thumb on the tick, or lock the helm there, to sail straight.
- If the tick turns red, the sails are asking for more rudder than exists. The boat will round up until you ease the main or drop the traveller.
- Sailing backwards, the rudder works in reverse.
- Stuck head to wind? Hold the rudder over; the boat keeps a little steerage and will turn slowly until the sails fill.

4 Instruments, telltales and hints

Tile	Meaning
Boat speed	Speed through the water in knots.
Target	The best speed this boat can reach on this wind angle in this wind, with perfect trim. Shows a dash when you point too close to the wind to sail.
Trim	Boat speed as a percentage of target. Green from 90 %, amber from 70 %, red below. It lags after a change because the boat needs time to accelerate.
TWS	True wind speed. Turns amber in a gust.
TWA	True wind angle: the angle between the bow and the true wind. S or P shows which side the wind is on.
AWA / AWS	Apparent wind angle and speed: the wind as felt on the moving boat. The sails are trimmed to this wind.
Heel	How far the boat leans, in degrees.
Helm	Rudder angle needed to sail straight. W is weather helm (the boat wants to turn into the wind), L is lee helm. Amber above 12°, red when it exceeds what the rudder can give.

Telltales

Telltales are short strips of yarn on the sail that show how the air flows over it. The panel shows four pairs: top and low on the genoa luff, top and low on the main leech.



Genoa telltales. The solid one is on the windward side of the sail, the dashed one on the leeward side, seen through the cloth. Green is the starboard side, red the port side.

State	What you see	Meaning	Fix
Luffing	Windward telltale lifts and flutters.	Sail too far out for the wind angle, or the boat is pointing too high.	Sheet in, or bear away.
Streaming	Both telltales fly straight aft.	Air flows cleanly on both sides.	Nothing. This is the goal.
Stalled	Leeward telltale droops and circles. On the main, the leech telltale curls behind the sail.	Sail too far in. The flow has separated on the leeward side.	Ease, or head up.
Flapping	Whole sail shakes.	The sheet is eased past the wind; the sail produces only drag.	Sheet in.
Hanging	Telltales hang limp.	Sailing deep downwind (apparent wind beyond about 115°). Normal.	Trim by boat speed.

When top and low disagree, the caption says so, for example “Top luffing”. That is a twist problem, solved with the genoa car, the vang or the mainsheet rather than by moving the whole sail (section 6).

Hints

The hint line picks the most urgent item, in roughly this order: too close to the wind, rounding up, a gust about to arrive, a whole sail luffing or stalled, a twist problem at the top of a sail, too much heel or helm, a wind shift, and finally small “fine tune” steps towards the best trim. Use it as a coach, then try to predict what it will say.

5 Wind, Best trim and settings

Automatic wind

In **Wind: auto** the wind blows from the direction and at the average strength chosen in Settings, and varies on its own:

- **Gusts** (off, light, moderate, strong) drift downwind as dark patches. Inside a patch the wind is stronger and slightly turned. The TWS tile turns amber and the hint warns a few seconds before a strong gust reaches you.
- **Wind shifts** (off, ±5°, ±12°) swing the wind direction slowly back and forth over one to three minutes. A shift that forces you to bear away is a *header*; one that lets you point higher is a *lift*.

Manual wind

In **Wind: manual** the wind is perfectly steady and does exactly what you tell it. Gusts and shifts are off. Use the panel under the gauges, or the keys:

- **Speed**, 2 to 30 knots, in steps of one knot.
- **From**, the compass direction the wind comes from, in steps of 5°.

Good experiments with manual wind:

- Hold your course and turn the wind aft in small steps. Watch the apparent wind follow and ease the sheets to keep the telltales streaming.

- Sail close-hauled and raise the wind from 8 to 20 knots. Note when heel and helm start to cost speed and how depowering brings it back.
- Turn the wind 10° ahead while close-hauled to feel a header, then 10° aft for a lift.

Best trim

Best trim calculates the fastest combination of main sheet, traveller, vang, genoa sheet and genoa car for your course and wind at the moment you press it. The controls then move there over a second or two so that you can see what changed.

- It is a one-off. After a course change or a wind change, the trim is no longer right; press again.
- It does not steer. Hold your course, or the trim will stop matching.
- Touching any sail control during the move cancels it.
- Pointing too close to the wind, no trim works, and the button tells you to bear away first.
- Where a control has no effect (for example the vang when the main is sheeted hard in), its position is arbitrary.

> **Use it to check, not to skip.** Trim by the telltales first, note your Trim %, then press Best trim and see which control you had furthest off.

Settings

Setting	Effect
Boat length	6 to 16 m. A longer boat is heavier, carries more sail and has a higher top speed. Smaller boats react faster.
Wind	Average true wind speed, 2 to 30 knots.
Wind direction	Where the wind comes from. Applies in both wind modes.
Wind control	Auto or manual, the same switch as the on-screen wind button.
Gusts, Wind shifts	Strength of the automatic variation. Inactive in manual mode.
Defaults	10 m boat, 14 knots from the north, light gusts, small shifts, automatic wind.

The simulation pauses while Settings is open.

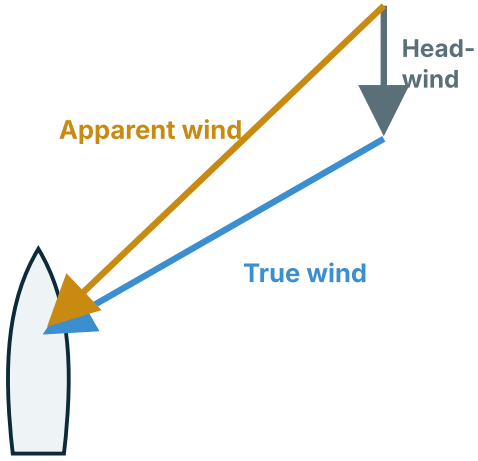
6 Basics of sail trimming

This section is a short primer. It uses the simulator's controls, but the ideas apply to any sloop.

True wind and apparent wind

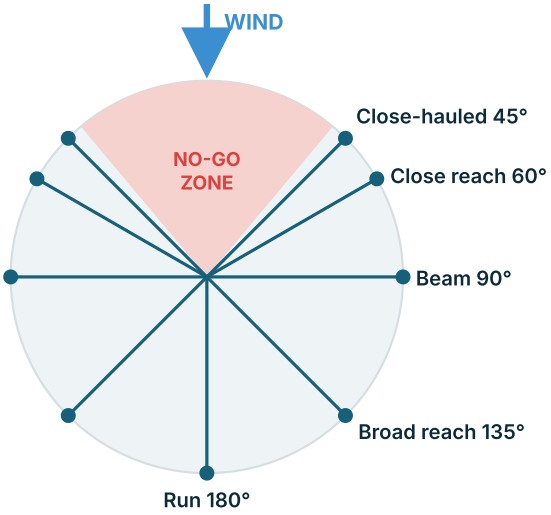
The **true wind** is what a flag on shore feels. A moving boat adds its own headwind, and the combination is the **apparent wind**. The sails only ever feel the apparent wind.

- Sailing upwind, the apparent wind is **stronger** than the true wind and comes from **further ahead**.
- Sailing downwind, it is **weaker**, because the boat runs away from it.
- When the boat speeds up, the apparent wind moves forward, so a boat that accelerates needs its sheets pulled in a little.



Apparent wind (amber) is the true wind (blue) plus the headwind the boat makes by moving (grey).

Points of sail



The angle between your course and the true wind (TWA) decides everything else.

- **No-go zone.** No boat sails straight into the wind. Inside roughly 40° the sails only flap.
- **Close-hauled.** As high as the boat will sail. Sheets hard in.
- **Reaching.** Wind from the side. The fastest courses.
- **Running.** Wind from behind. Sails right out.

The basic rule for the sheets follows directly: **the further the wind is aft, the further out the sails go.**

The golden rule: ease until it luffs, then sheet in

A sail works like a wing. It makes the most drive at a moderate angle to the apparent wind, about 15° to 25°. Too small an angle and it luffs: little power. Too large and the flow breaks away on the leeward side: it stalls, power drops and drag climbs.

A stalled sail looks full and quiet, which is why oversheeting is the most common trimming mistake. A luffing sail flutters visibly. So, when unsure:

> **When in doubt, let it out.** Ease the sheet until the windward telltale just starts to lift, then sheet in until it streams again. Repeat after every change of course or wind.

The telltales turn this into two simple reflexes. The telltale that misbehaves tells you which way to go:

- **Windward telltale lifts:** move the sail **towards** it (sheet in), or steer away from it (bear away).
- **Leeward telltale droops:** ease the sail **towards** it, or steer towards the wind (head up).

On a reach you hold the course and trim the sails. Close-hauled the sheets are already in, so you do the opposite: leave the sails and **steer** to keep the genoa telltales streaming.

Genoa first, then the main

The genoa meets clean wind and bends it before it reaches the mainsail. The main therefore lives in wind that comes from further ahead and has to be sheeted **closer to the centreline than the genoa**. Trim in this order:

1. **Genoa sheet** until the low telltales stream.
2. **Genoa car** until the top telltales break at the same moment as the low ones.
3. **Main sheet and traveller** until the main leech telltales stream. If the front of the main bubbles, the genoa is backwinding it: sheet the main in a little or ease the genoa.
4. **Twist** of the main with the mainsheet (upwind) or the vang (off the wind).
5. **Balance.** Check heel and helm, and depower if either is large.

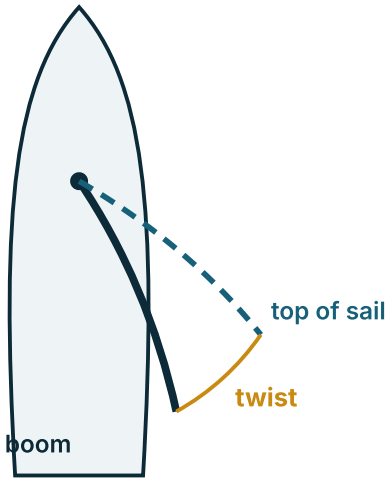
An oversheeted genoa also chokes the gap between the sails and robs the main of power, so getting the genoa right pays twice.

Twist

Twist is the difference in angle between the bottom and the top of a sail. Some twist is correct: the wind is stronger high above the water, so the apparent wind aloft comes from further aft and the top of the sail should be a little more open than the bottom.

- **Too much twist:** the top luffs while the bottom works. Power is lost.
- **Too little twist:** the top stalls while the bottom streams. The leech is “closed”.
- **Just right:** top and low telltales behave the same.

Twist is also the gentlest way to reduce power in a breeze, because spilling wind from the top of the sail reduces heel the most.



Mainsail from above. The solid line is the boom, the dashed line the upper part of the sail, as drawn in the simulator.

Control	Less twist (close the leech)	More twist (open the leech)	When it matters
Main sheet	Sheet hard in	Ease	Upwind, in the last part of the sheet's travel. Further out it mainly changes the boom angle.
Vang	Tighten	Ease	Reaching and running, when the eased sheet no longer holds the boom down.
Genoa car	Forward	Aft	Always. Move it forward as the genoa sheet is eased.
Traveller	Changes the boom angle and leaves twist alone.		Upwind: fine adjustment of power and helm without touching the leech.

Upwind, mainsheet and traveller work as a pair: the **sheet sets the twist**, the **traveller sets the angle**.

Heel, helm and depowering

Wind force on the sails pushes the boat sideways and heels it. A little heel is unavoidable; a lot is slow. The sails lose effective area, the hull drags, the keel and rudder work less well, and the boat tries to turn into the wind (**weather helm**). Holding it straight needs a lot of rudder, and a rudder held at a large angle is a brake.

- A few degrees of weather helm are normal and healthy.
- The main sits behind the keel and turns the boat **into** the wind; the genoa sits ahead of it and turns the boat **away**. Easing the main reduces weather helm; easing the genoa increases it.
- If the boat rounds up and will not answer the rudder, ease the main at once.

When heel passes about 25° or the helm gets heavy, take power out, in this order:

1. **Traveller down.** Quick, and the leech stays as it was.
2. **More twist.** Ease the mainsheet slightly, ease the vang, move the genoa car aft.
3. **Ease the main** further, even until its front luffs a little. A boat sailing flatter with a slightly luffing main is faster than one lying on its side.

On a real boat the next step is to reef. The simulator has no reefs, so above about 20 knots depowering is all you have.

Sailing downwind

- Ease the main until the boom is nearly square to the wind, and tighten the vang so the top of the sail does not twist away.
- The sails now work by simply catching wind rather than as wings. The telltales hang limp, and that is normal. Trim by boat speed.
- Dead downwind the main shades the genoa. Heading up 10° to 20° fills the genoa again and is usually faster.
- The apparent wind is weak downwind, so it feels calm. Remember that when you turn back upwind in a breeze.

Trim by point of sail

Course	Genoa	Main	Watch
Close-hauled	Sheet hard in. Car set so top and low telltales break together.	Sheet nearly hard in, traveller around the middle or slightly up in light wind. Vang has no effect.	Steer to the genoa telltales. Do not pinch: a luffing genoa with the sheet hard in means bear away.
Close reach	Ease until the telltales stream. Car forward a little.	Ease sheet, traveller down. Start using the vang.	Heel and weather helm in a breeze.
Beam reach	Ease well out, car forward.	Ease sheet, vang on to keep the top working.	Top telltales: twist is the usual problem here.
Broad reach	Eased far out. It begins to stall whatever you do.	Boom well out, vang tight.	Boat speed more than telltales.
Run	Right out, partly blanketed by the main.	Boom nearly square, vang tight.	Head up a little to fill the genoa.

Troubleshooting

Symptom	Likely cause	Try
Slow, sails look full and quiet	Oversheeted and stalled.	Ease both sheets until the telltales lift, then sheet in slightly.
Genoa windward telltale lifts	Genoa too far out, or pointing too high.	Sheet in or bear away.
Genoa leeward telltale droops	Genoa too far in.	Ease or head up.
Top genoa telltale lifts first	Too much twist.	Car forward.
Top genoa leeward telltale stalls first	Leech too tight.	Car aft.
Main leech telltales hide behind the sail	Main oversheeted.	Traveller down or ease the sheet.
Only the top main telltale hides	Leech too closed.	Ease the mainsheet a touch (traveller up), or ease the vang.
Top of the main luffs	Too much twist.	Upwind: more sheet, traveller down. Reaching: tighten the vang.
Heavy helm, boat rounds up	Too much power in the main, too much heel.	Traveller down, then ease the main.
Boat stopped, sails flapping	In the no-go zone.	Hold the rudder over until the sails fill, then build speed before heading up.
Slides sideways after a tack	Too slow for the keel to grip.	Bear away, ease slightly, gain speed, then point up again.

7 How the simulator works

SailTrim aims for correct behaviour and trends, not engineering accuracy. If a sailor would say “yes, that is what the boat does”, the model is doing its job. Do not read the numbers as predictions for a particular real boat.

The loop

Sixty times a second the simulator works out the apparent wind from the true wind and the boat's motion, the force on each sail, the resistance of the hull, the side force of the keel, the heel and the turning effect, and moves the boat accordingly.

Sails

- **Sheet to angle.** A sheet sets the furthest a sail can swing out. If the sheet allows more than the wind angle, the sail lines up with the wind and flaps, giving drag and no drive.
- **Angle of attack** is the apparent wind angle minus the sail angle. Lift and drag are read from a curve: lift rises to a peak near 24°, then falls while drag keeps climbing. Below 10° the telltales show luffing, above 28° stalled.
- **Force.** Sail force grows with sail area and with the *square* of the apparent wind speed. Twice the wind gives four times the force. Lift acts across the apparent wind and drag along it; the part pointing forward is drive, the part pointing sideways causes heel and leeway.
- **Two sections per sail.** Each sail is split into a lower part (70 % of the area) and an upper part (30 %). The upper part feels about 15 % more true wind than the lower and sits at the lower angle plus the twist. This is why a little twist is fastest and why top and low telltales can disagree.
- **Twist ranges.** Main: 2° with the sheet hard in, opening to 24° as the sheet is eased, unless the vang holds it tighter (down to 5° with the vang fully on). Genoa: 6°, plus or minus 6° from the car, plus up to 16° more as its sheet is eased.
- **Slot effect.** A working genoa turns the wind reaching the main by up to 6°, so the main needs a tighter angle. A stalled genoa disturbs the main and costs it 20 % of its lift.
- **Blanketing.** With the apparent wind beyond 150° the main increasingly shades the genoa, down to 30 % of its force dead downwind.

Hull, keel and speed

- **Resistance** grows with the square of speed and then very steeply near **hull speed**, which depends on waterline length: about 2.4 × √(waterline in metres) knots, 7.2 knots for the 10 m boat. More wind beyond that adds heel, not much speed. Heel adds further drag.
- **Keel.** The keel resists sideways motion, but it only produces force when water flows over it at a small angle (leeway), and that force grows with the square of boat speed. A slow boat therefore slides sideways; build speed first, then point.
- **Heel** settles where the heeling effect of the sails (side force times its height) equals the righting effect of the hull and ballast. It is limited to 45°.

Helm balance

The main pushes sideways behind the keel and the genoa ahead of it, so their balance turns the boat. Heel moves the drive force out to leeward and adds weather helm. The **Helm** tile and the green tick show

the rudder angle that cancels this. Rudder angle costs speed, a heeled rudder grips less, and if more than the full 30° would be needed the boat rounds up.

Target speed and Trim %

For the present wind angle and strength the simulator searches all combinations of the five sail controls for the highest steady speed, rejecting trims that need more than 80 % of full rudder. That speed is the **Target**, and Trim % is your speed divided by it. The table of targets is filled in gradually while you sail, at steps of 2° of wind angle and about one knot of wind. **Best trim** runs the same search for your exact situation.

The boat

10 m reference boat	Value	Changes with length
Waterline length	8.8 m	in proportion
Displacement	5 000 kg	with the cube of length
Mainsail / genoa (135 %)	28 m² / 34 m²	with the square of length
Keel area	1.6 m²	with the square of length

What is not modelled

- Waves, current, and crew weight on the rail.
- Reefing, spinnakers and sail changes.
- Sail depth and draft controls: outhaul, cunningham, backstay, halyard tension.
- Tacking technique: the genoa changes side by itself.